Written HW 5 — Full Solutions

Problem statements from the hand-written worksheet (not a textbook exercise set): Ohm's law, the junction rule, the loop rule, and series & parallel resistors. Every battery is ideal ($r = 0$). Problems 1–9 are Section 1 (required, 2 pts each); EC1–EC9 are Section 2 (extra credit, 1 pt each).

1 (2 pts)

A resistor has resistance $R = 25\ \Omega$. A voltmeter placed across it reads $\Delta V = 15$ V. What current flows through the resistor?

MODEL: Ohm's law relates the current, resistance, and voltage drop of a single resistor.

SOLVE:

$$I = \frac{\Delta V}{R} = \frac{15\text{ V}}{25\ \Omega} = \boxed{0.60\text{ A}}.$$

2 (2 pts)

Three wires meet at a junction: 2.0 A and 3.5 A flow in, and a current I flows out. Find I.

MODEL: Kirchhoff's junction rule: charge doesn't pile up at a junction, so the total current in equals the total current out.

SOLVE:

$$I = 2.0\ \text{A} + 3.5\ \text{A} = \boxed{5.5\ \text{A}}.$$

3 (2 pts)

Four wires meet at a junction: 4.0 A and 1.0 A flow in, 2.0 A and I flow out. Find I.

MODEL: Junction rule again, now with currents on both sides of the balance.

SOLVE:

$$4.0\text{ A} + 1.0\text{ A} = 2.0\text{ A} + I \;\Rightarrow\; I = \boxed{3.0\text{ A}}.$$

4 (2 pts)

A single-loop circuit has a battery with $\varepsilon = 20$ V and two resistors in series. A voltmeter reads $\Delta V_1 = 12$ V across the first resistor. Find ΔV_2 across the second.

MODEL: Kirchhoff's loop rule: going all the way around a loop, the total voltage gained equals the total voltage lost. For two series resistors, their drops must add up to the full emf.

SOLVE:

$$\varepsilon = \Delta V_1 + \Delta V_2 \;\Rightarrow\; \Delta V_2 = 20 \text{ V} - 12 \text{ V} = \boxed{8.0 \text{ V}}.$$

5 (2 pts)

Same idea, different numbers: a single loop has $\varepsilon = 24$ V and two series resistors, with $\Delta V_1 = 9.0$ V. Find ΔV_2.

MODEL: Same loop rule as Problem 4.

SOLVE:

$$\Delta V_2 = \varepsilon - \Delta V_1 = 24 \text{ V} - 9.0 \text{ V} = \boxed{15 \text{ V}}.$$

6 (2 pts)

Two resistors, $R_1 = 4.0\ \Omega$ and $R_2 = 8.0\ \Omega$, are in series with a battery of $\varepsilon = 18$ V. Find (a) the equivalent resistance, and (b) the current supplied by the battery.

MODEL: Series resistors add directly; the battery then drives that single equivalent resistance.

SOLVE: (a)

$$R_{\text{eq}} = R_1 + R_2 = 4.0\ \Omega + 8.0\ \Omega = \boxed{12\ \Omega}.$$

(b)

$$I = \frac{\varepsilon}{R_{\text{eq}}} = \frac{18\ \text{V}}{12\ \Omega} = \boxed{1.5\ \text{A}}.$$

7 (2 pts)

Three resistors, $R_1 = 2.0\ \Omega$, $R_2 = 3.0\ \Omega$, and $R_3 = 5.0\ \Omega$, are in series with a battery of $\varepsilon = 20$ V. Find (a) the equivalent resistance, and (b) the current supplied by the battery.

MODEL: Same idea as Problem 6, now with three series resistors.

SOLVE: (a)

$$R_{\text{eq}} = 2.0\ \Omega + 3.0\ \Omega + 5.0\ \Omega = \boxed{10\ \Omega}.$$

(b)

$$I = \frac{\varepsilon}{R_{\text{eq}}} = \frac{20\text{ V}}{10\ \Omega} = \boxed{2.0\text{ A}}.$$

8 (2 pts)

Two resistors, $R_1 = 6.0\ \Omega$ and $R_2 = 3.0\ \Omega$, are in parallel across a battery of $\varepsilon = 12$ V. Find (a) the equivalent resistance, and (b) the total current from the battery.

MODEL: Parallel resistors combine by reciprocals; both branches see the full battery voltage.

SOLVE: (a)

$$R_{\text{eq}} = \left(\frac{1}{6.0\ \Omega} + \frac{1}{3.0\ \Omega}\right)^{-1} = \boxed{2.0\ \Omega}.$$

(b)

$$I_{\text{tot}} = \frac{\varepsilon}{R_{\text{eq}}} = \frac{12\ \text{V}}{2.0\ \Omega} = \boxed{6.0\ \text{A}}.$$

REVIEW: Check directly: $I_1 = \varepsilon/R_1 = 12/6.0 = 2.0$ A and $I_2 = \varepsilon/R_2 = 12/3.0 = 4.0$ A, so $I_{\text{tot}} = I_1 + I_2 = 6.0$ A — matches.

9 (2 pts)

Two resistors are in parallel across a battery of $\varepsilon = 12$ V. $R_1 = 6.0\ \Omega$, and the *total* current measured from the battery is 5.0 A. Find (a) the current through R_2, and (b) the value of R_2.

MODEL: In parallel, each branch independently sees the full battery voltage, so R_1's current can be found on its own; the junction rule then gives whatever's left for R_2.

SOLVE: (a) R_1 alone carries

$$I_1 = \frac{\varepsilon}{R_1} = \frac{12\text{ V}}{6.0\ \Omega} = 2.0\text{ A},$$

so by the junction rule,

$$I_2 = I_{\text{tot}} - I_1 = 5.0\text{ A} - 2.0\text{ A} = \boxed{3.0\text{ A}}.$$

(b)

$$R_2 = \frac{\varepsilon}{I_2} = \frac{12\text{ V}}{3.0\text{ A}} = \boxed{4.0\ \Omega}.$$

EC1 (1 pts)

A current of 0.40 A flows through a resistor when a voltmeter across it reads 8.0 V. Find R.

SOLVE:

$$R = \frac{\Delta V}{I} = \frac{8.0 \text{ V}}{0.40 \text{ A}} = \boxed{20 \ \Omega}.$$

EC2 (1 pts)

At a junction, 6.0 A flows in and splits into two outgoing wires; one carries 2.5 A. Find the other.

SOLVE:

$$I = 6.0\ \text{A} - 2.5\ \text{A} = \boxed{3.5\ \text{A}}.$$

EC3 (1 pts)

Three wires carry current *into* a junction: 1.5 A, 2.0 A, and 2.5 A. One wire carries it all back out. Find that current.

SOLVE:

$$I = 1.5\text{ A} + 2.0\text{ A} + 2.5\text{ A} = \boxed{6.0\text{ A}}.$$

EC4 (1 pts)

A single loop has $\varepsilon = 9.0$ V and *three* series resistors. Voltmeters across two of them read 2.0 V and 3.0 V. Find the reading across the third.

SOLVE:

$$\Delta V_3 = \varepsilon - \Delta V_1 - \Delta V_2 = 9.0 \text{ V} - 2.0 \text{ V} - 3.0 \text{ V} = \boxed{4.0 \text{ V}}.$$

EC5 (1 pts)

A single loop has $\varepsilon = 30$ V and three series resistors. $\Delta V_1 = 10$ V and $\Delta V_2 = 8.0$ V. Find ΔV_3.

SOLVE:

$$\Delta V_3 = 30 \text{ V} - 10 \text{ V} - 8.0 \text{ V} = \boxed{12 \text{ V}}.$$

EC6 (1 pts)

Two resistors, $R_1 = 3.0\ \Omega$ and $R_2 = 7.0\ \Omega$, are in series with a battery of $\varepsilon = 20$ V. Find the current supplied by the battery.

SOLVE:

$$I = \frac{\varepsilon}{R_1 + R_2} = \frac{20\text{ V}}{10\ \Omega} = \boxed{2.0\text{ A}}.$$

EC7 (1 pts)

Two resistors are in series with a battery of $\varepsilon = 15$ V. $R_1 = 2.0\ \Omega$, and the current measured in the circuit is 1.5 A. Find R_2.

SOLVE: The same current flows through both series resistors, so the equivalent resistance is

$$R_{\text{eq}} = \frac{\varepsilon}{I} = \frac{15\text{ V}}{1.5\text{ A}} = 10\ \Omega \ \Rightarrow\ R_2 = R_{\text{eq}} - R_1 = 10\ \Omega - 2.0\ \Omega = \boxed{8.0\ \Omega}.$$

EC8 (1 pts)

Two resistors, $R_1 = 8.0\ \Omega$ and $R_2 = 24\ \Omega$, are in parallel across a battery of $\varepsilon = 24$ V. Find the total current supplied by the battery.

SOLVE: Each branch sees the full 24 V independently:

$$I_1 = \frac{24\text{ V}}{8.0\ \Omega} = 3.0\text{ A}, \qquad I_2 = \frac{24\text{ V}}{24\ \Omega} = 1.0\text{ A} \;\Rightarrow\; I_{\text{tot}} = I_1 + I_2 = \boxed{4.0\text{ A}}.$$

EC9 (1 pts)

Two resistors, $R_1 = 5.0\ \Omega$ and $R_2 = 20\ \Omega$, are in parallel across a battery. The *total* current from the battery is measured to be 5.0 A. Find ε.

SOLVE: Both branches share the same unknown voltage ε, so

$$I_{\text{tot}} = \frac{\varepsilon}{R_1} + \frac{\varepsilon}{R_2} = \varepsilon\left(\frac{1}{5.0\ \Omega} + \frac{1}{20\ \Omega}\right) = \varepsilon\left(\frac{1}{4.0\ \Omega}\right)$$

$$\Rightarrow\ \varepsilon = (5.0\ \text{A})(4.0\ \Omega) = \boxed{20\ \text{V}}.$$